MAE 575 Coding Project

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4/2/2026

1.) Reproduce figures five through seven using hydrogen fuel.

Figure five shows properties for a ϕ of 0.5. The flow remains attached throughout the combustor in this case. The paper calculated the maximum P/P2 = 2.19, maximum T/T2 = 2.61, and minimum Mach number of 1.48. The reproduction script returns values of 2.28, 2.62, and 1.47, respectively. The pressure gradient is slightly too steep in the isolator, which results in the skewed P/P2, but overall, this is a very faithful reproduction.

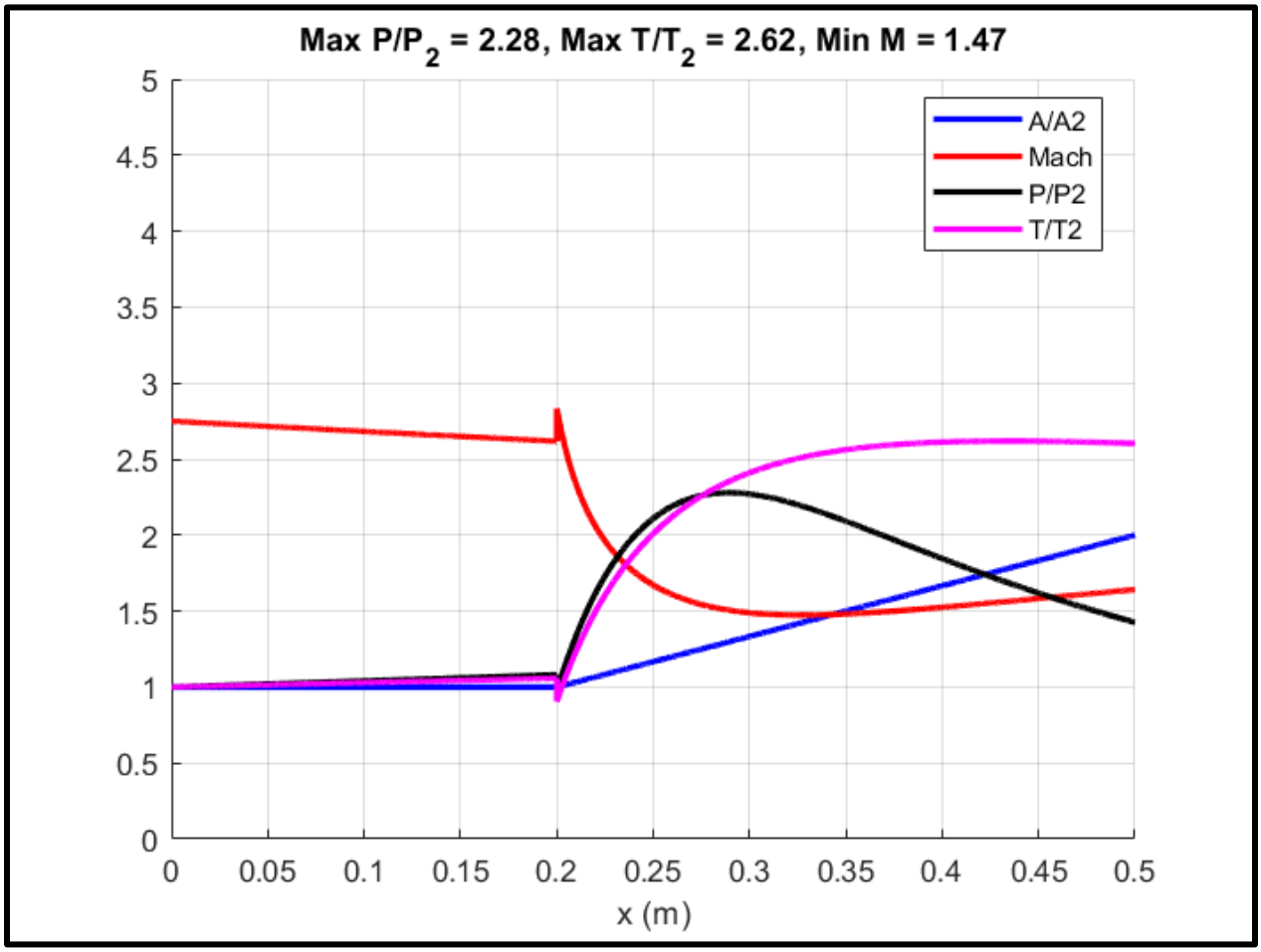

Figure 1: $\phi = 0.5$ Case, Hydrogen

In figure six, the ϕ is raised to 0.72 which causes separation in the isolator. The separation point is given as 0.180 meters and the flow reattaches at x = 0.213. The paper gives the maximum P/P2 as 3.51 & the minimum Mach as 1.087. In my analysis, due to the slightly steeper Mach decrease in the isolator, I increased the initial Mach number to 2.95 to more accurately replicate the figure. This change was necessary since the Mach curve comes so close to one, which causes the differential equations to blow up. With that change, the maximum P/P2 returned by the script is 3.40 & The minimum Mach number is 1.23. In this

case, the maximum pressure is slightly low and the minimum Mach number is slightly high, but the shape is faithfully recreated.

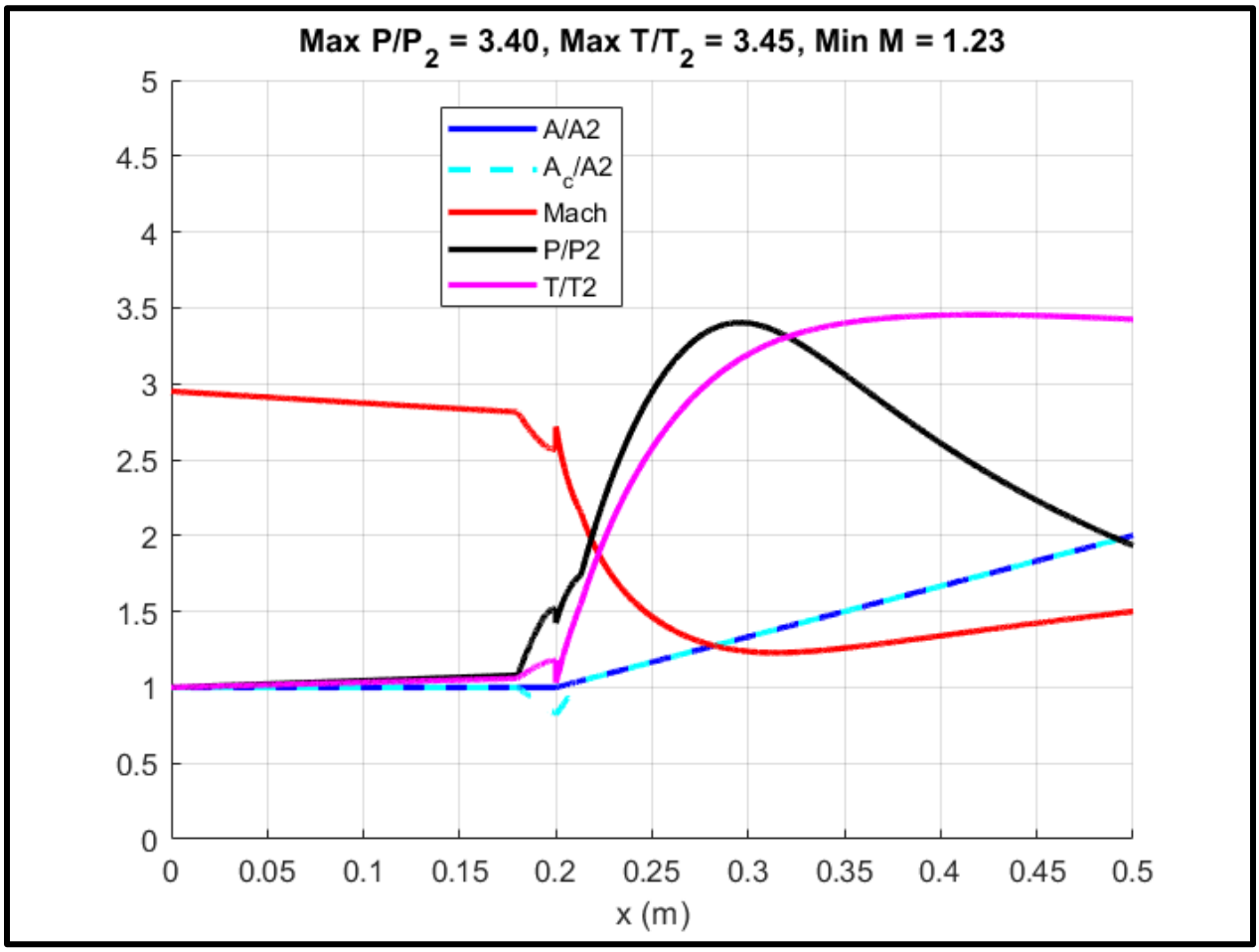

Figure 2: $\phi = 0.72$ *Case, Hydrogen Fuel*

Figure seven increases the ϕ to 0.81, which causes the Mach number to drop below one in a 'thermal throat.' The minimum Mach number is 0.96 at the point of reattachment, and the maximum P/P2 = 4.24. I struggled to recreate this figure, the way the differential equations are formulated, A/A_c must be prescribed, and I struggled to create the same profile as the paper. Furthermore, the isolator pressure ratio did not increase as quickly as expected when using the provided equations for separated isolator flows. Regardless, I was able to capture the spirit of figure seven, matching the minimum Mach number (since it was given I forced the solution to match,) and achieving a maximum P/P2 of 4.36.

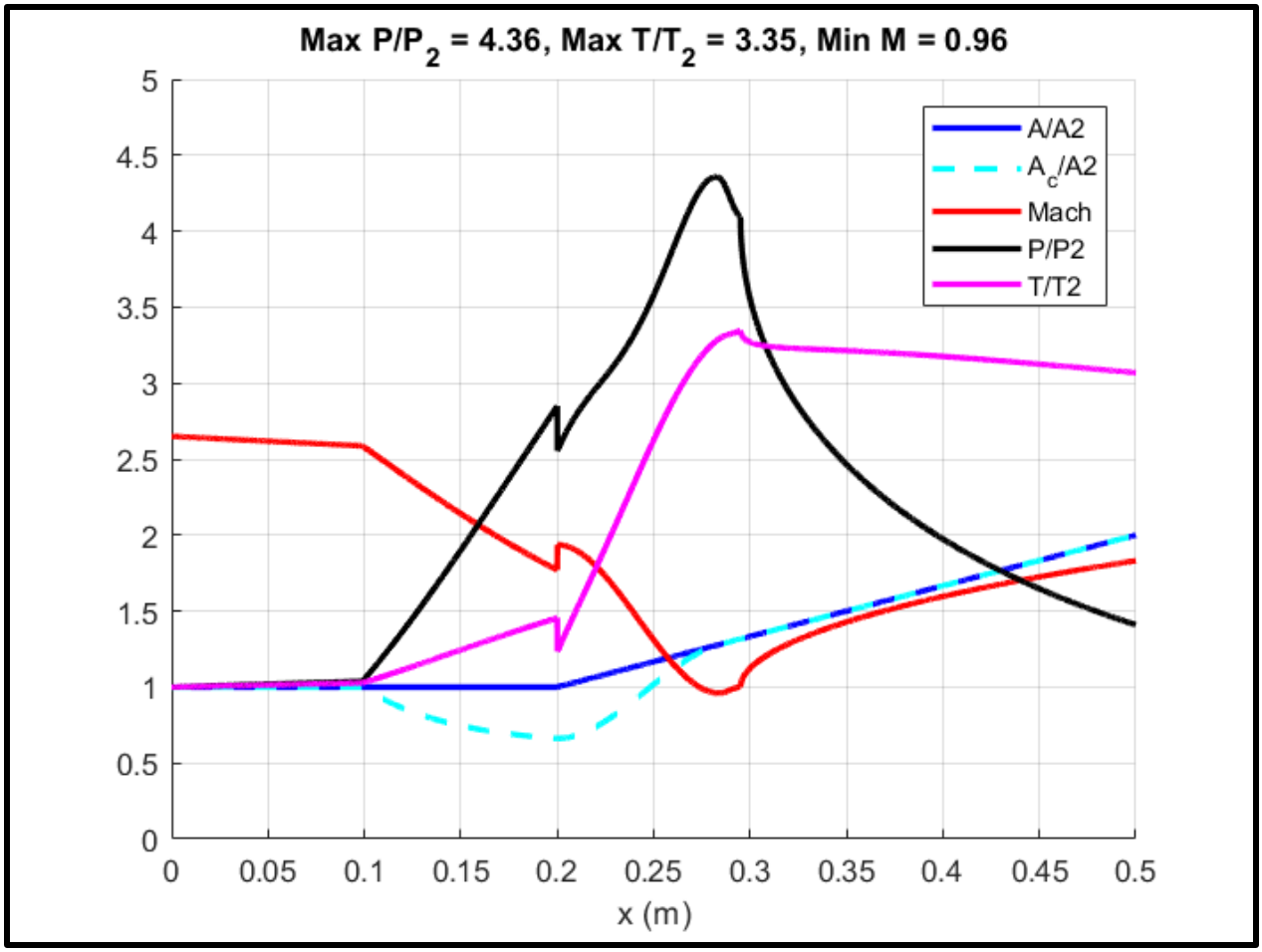

Figure 3: $\phi = 0.81$ *Case, Hydrogen Fuel*

2.) The next task is to recreate the above figures using JP8 fuel. The main changes here are the heat of combustion change ($h_{pr} = 45\ MJ/kg$) and stoichiometric fuel air ratio ($f_{st} = 0.685.$) The fuel choice will not affect point of separation or area curve.

For figure five, everything will stay attached, so we just update the code with the new h_{pr} & f_{st} then rerun. The results look quite similar, with slightly lower values for max pressure, temperature, and minimum Mach number.

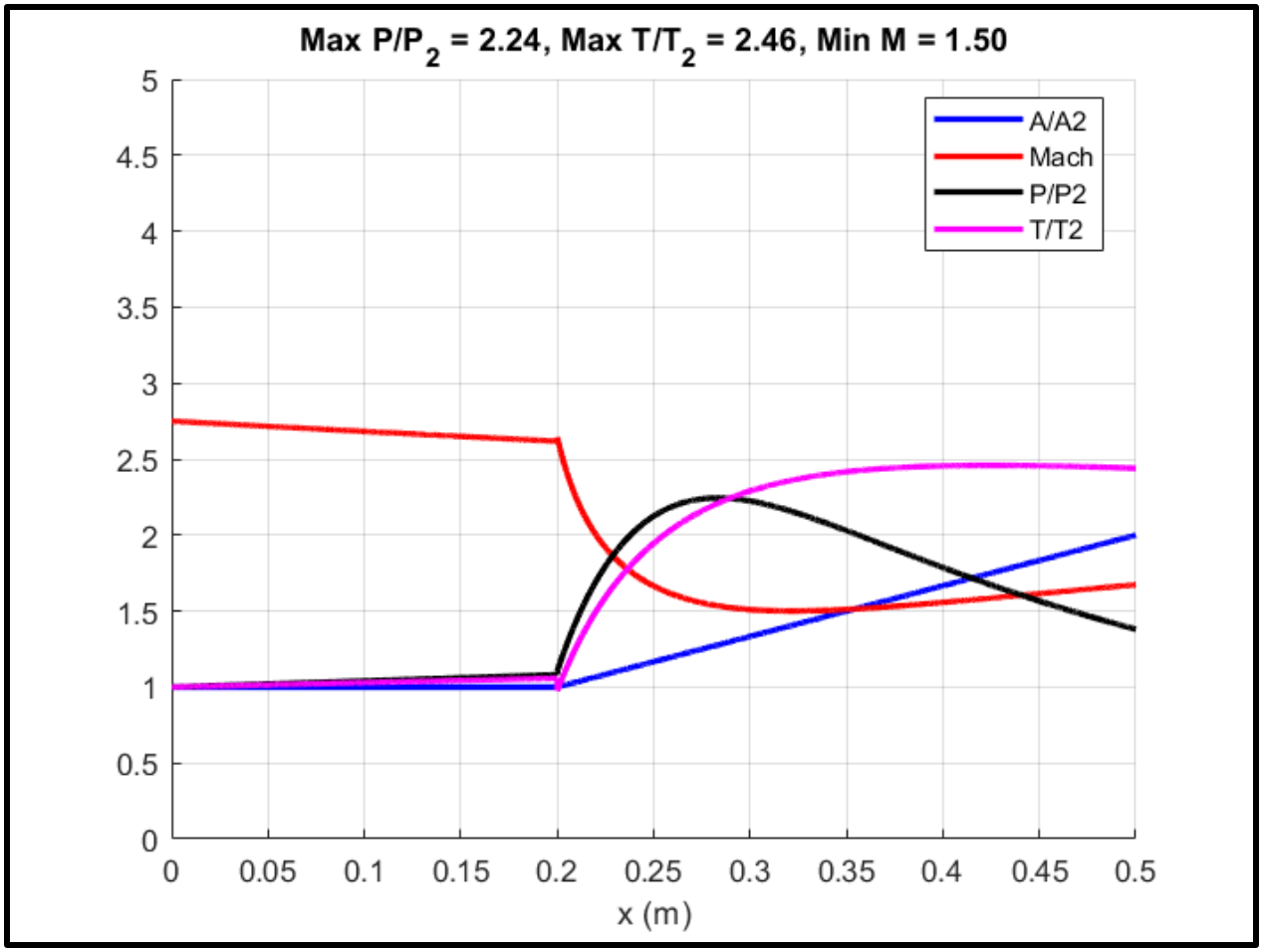

Figure 4: ϕ=0.5 Case, JP8

For figure six, the flow detaches upstream of the combustor & reattaches in the burner. The results look very similar to the hydrogen case, with a lower max pressure, lower maximum temperature, and slightly higher minimum Mach number. This matches assumed behavior for a fuel with a much lower heat of combustion.

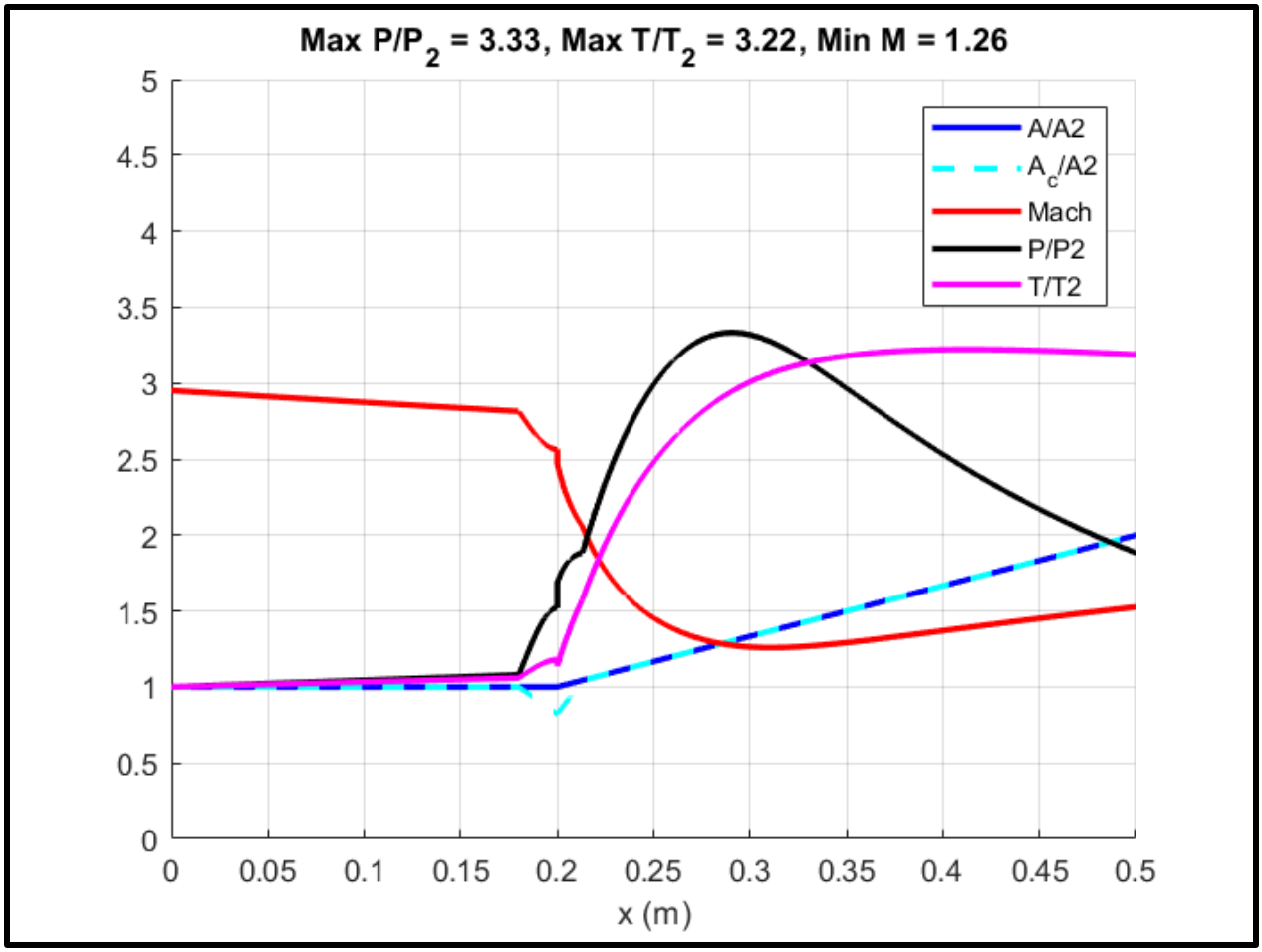

Figure 5: ϕ = 0.72 Case, JP8

The figure seven reproduction suffers from the same deficiencies as the original. The area curve is an inexact reproduction that has been forced into an approximate shape, and the isolator's slopes are too mild. The pressure & Mach curves are quite similar, as my script prescribes the area curve, forcing them to match up. The maximum temperature is quite a bit lower, as expected.

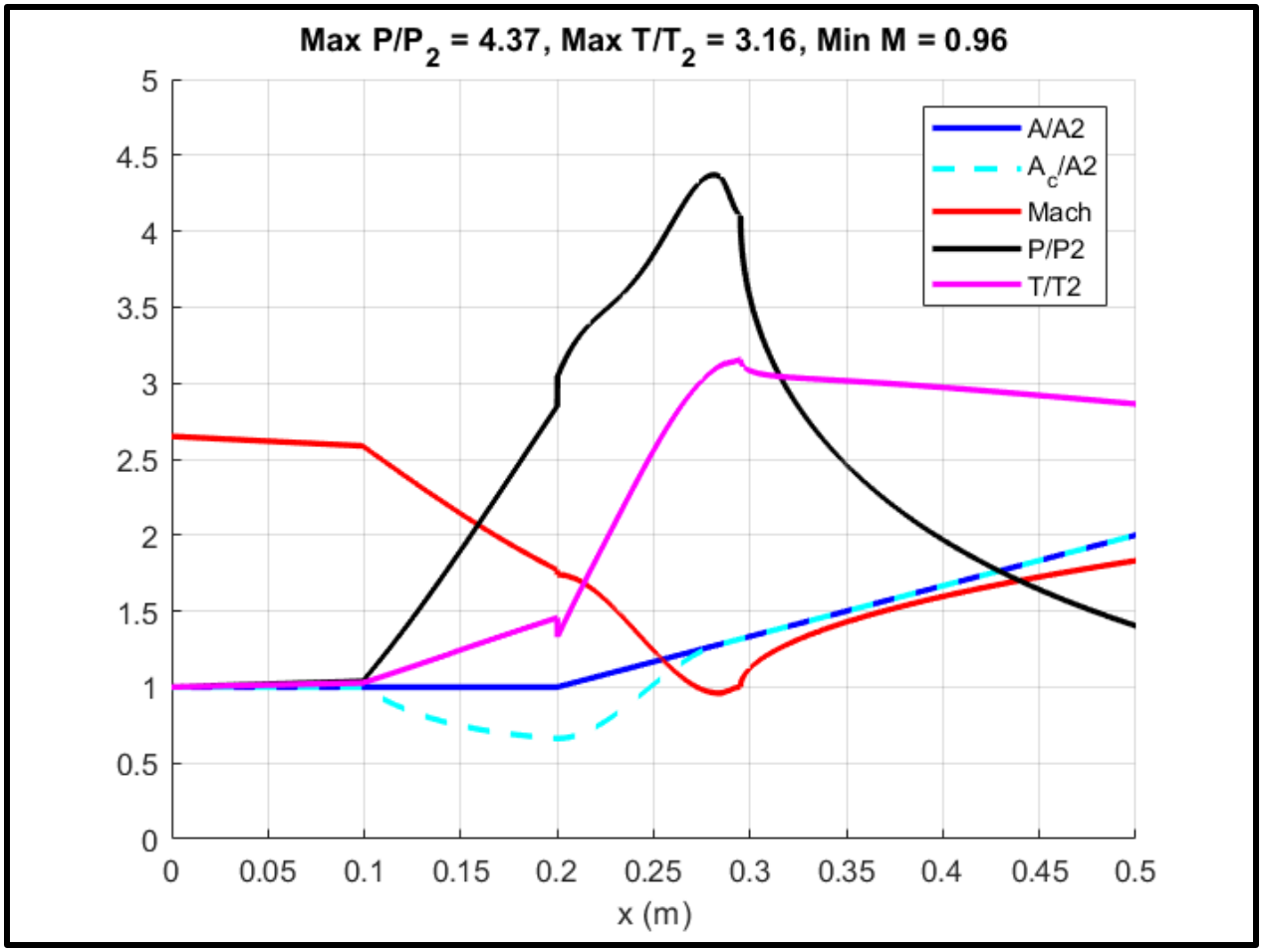

Figure 6: ϕ = 0.81 Case, JP8

3.) To determine the onset of thermal choking with JP-8, we will sweep the phi across known values until we observe thermal choking, defined as Mach crossing into the subsonic region. We know that thermal choking is not present for a $\phi = 0.72$, and that it is present for 0.81. We can check various values in that region until we find the approximate onset point. The separation point and attachment point will be treated as linearly varying between the endpoints.

$$[sep, attach]\phi = 0.72 = [0.180, 0.213]$$

$$[sep, attach]\phi = 0.81 = [0.099, 0.284]$$

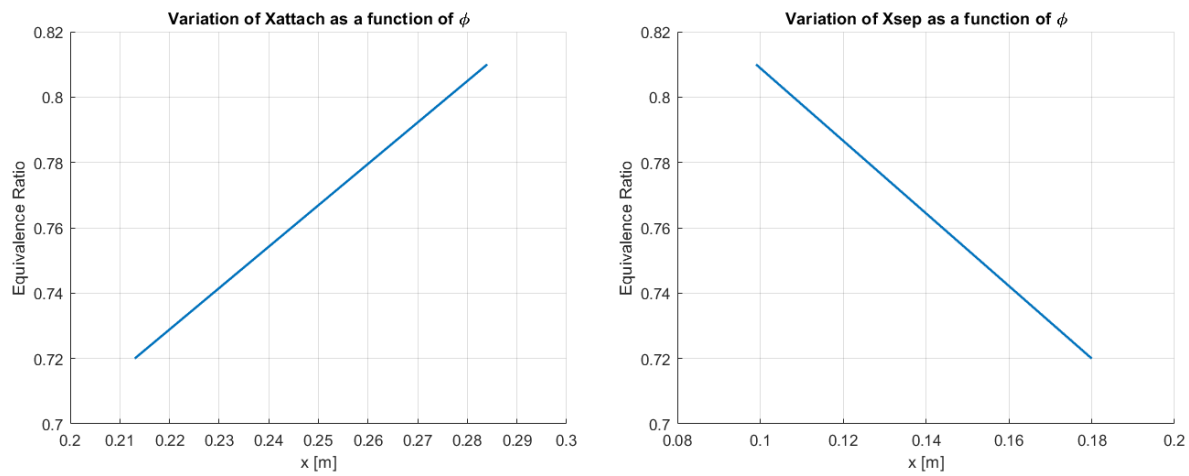

Figure 7: Variation of Xsep & Xattach

Using this method, I found the onset of the thermal throat to come between a phi of 0.73 & 0.74, with an xsep of 0.153 meters and an xattach of 0.2367 meters.

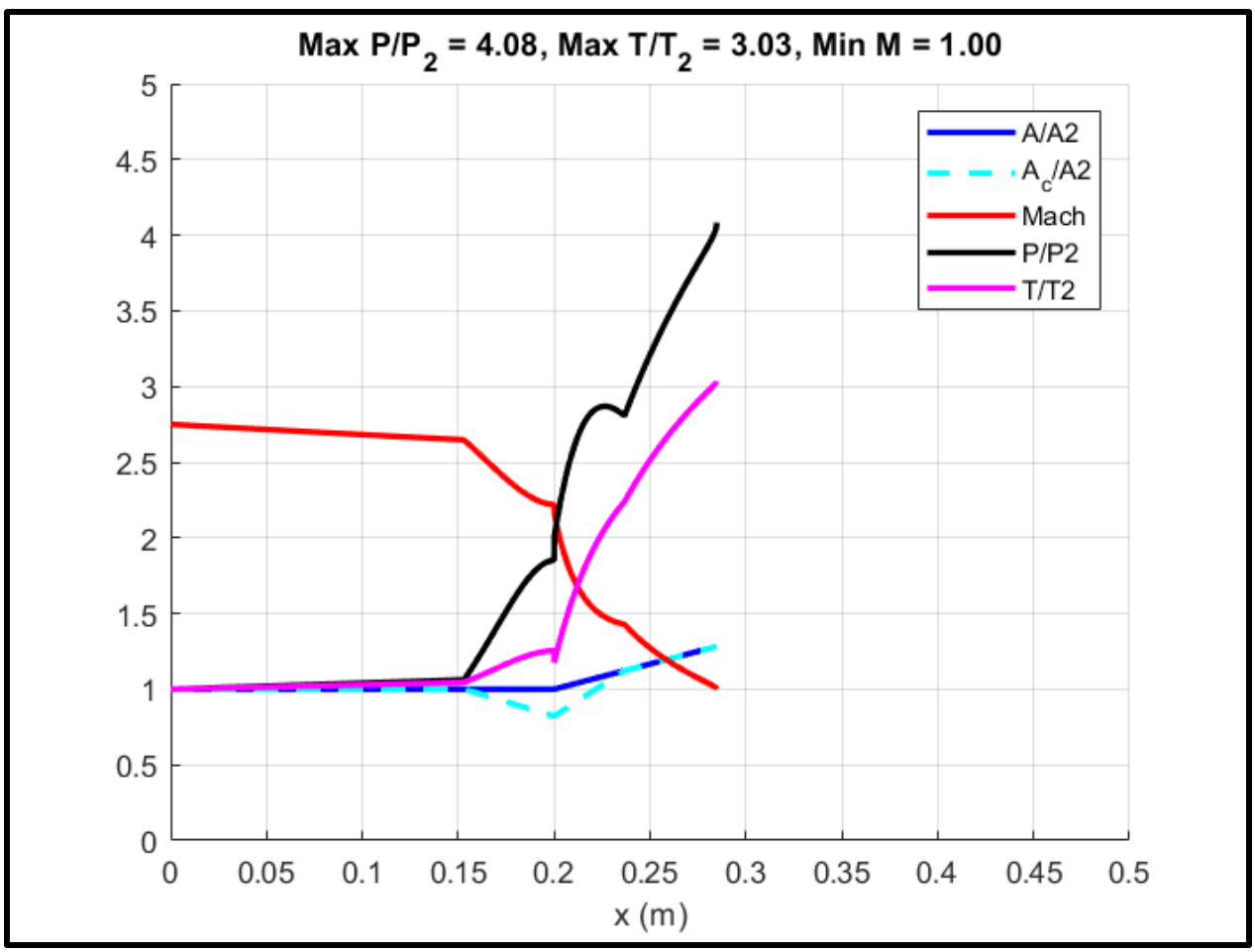

Figure 8: $\phi = 0.74$ case, JP8